

RESEARCH ARTICLE

DPF-Bi-RRT*: An Improved Path Planning Algorithm for Complex 3D Environments With Adaptive Sampling and Dual Potential Field Strategy

LIN GE^{1,2}, SWEE KING PHANG², (Senior Member, IEEE), AND NOHAIDDA SARIFF²

¹School of Urban Construction, Hangzhou Polytechnic, Hangzhou 310000, China

²School of Engineering, Taylor's University, Subang Jaya, Selangor 47500, Malaysia

Corresponding author: Swee King Phang (sweeking.phang@taylors.edu.my)

The work of Swee King Phang was supported by Taylor's University, Malaysia, through the Taylor's Internal Research Grant Scheme—Impact Lab Grant TIRGS-ILG/1/2023/SOE/001.

ABSTRACT This research work, we introduce a path planning algorithm called DPF-Bi-RRT* which integrates a Dual Potential Field mechanism with a Targeted Sampling Strategy to address path planning issues in complex 3D spaces. The algorithm achieves a good trade off between the global path optimization and precise local obstacle avoidance by combining dual-attraction and dual-repulsion mechanisms. The algorithm achieves effective exploration of path regions of high quality by dynamically adjusting the sampling distribution. Additionally, a biased random sampling strategy improves computational efficiency by directing sampling resources toward sections with higher promise of optimal paths, dramatically reducing computational cost. The dual potential field model lead to more flexible method in collision avoidance and can improve the precision of collision avoidance, especially in cluttered dynamical spaces. We perform comparative simulations of DPF-Bi-RRT* versus RRT*, Bi-RRT*, and APF-Bi-RRT* across three defined environments to demonstrate that DPF-Bi-RRT* results in lower average node counts, less computational time, and longer path lengths than all three. Results validate its capability of generating smooth, collision free and globally optimized paths making it especially suitable for autonomous aerial vehicle (AAV) navigation in complex 3D environments.

INDEX TERMS adaptive sampling strategy, biased random sampling, collision avoidance, complex environments, dual potential fields, global and local optimization, path planning, AAV.

I. INTRODUCTION

Due to their high mobility and operational flexibility, Autonomous Aerial Vehicles (AAVs) have recently become a core tool in many areas, such as disaster response, traffic monitoring, logistics and transport, military reconnaissance and precision agriculture [1], [2], and generating collision-free and efficient flight paths in complex environments is a core technology for realizing autonomous AAV capabilities [3]. AAVs remain intending to perform basic fast,

flexible navigation with efficient completion of tasks, and rational use of resources [4]. Nevertheless, reaching these goals in scenarios with complex environments and dense obstacles is still a substantially challenging task; making algorithms able to work in the tradeoff between running speed, adaptability, and stability [5].

Sampling-based algorithms like Probabilistic Roadmap (PRM) construct a graph of feasible paths by connecting randomly sampled configurations with collision-free edges. These methods perform well in static environments and high-dimensional spaces but often struggle to adapt to dynamic or rapidly changing scenarios due to their reliance

The associate editor coordinating the review of this manuscript and approving it for publication was Zhenbao Liu¹.

on pre-computed roadmaps [6], [7], [8]. Optimization-based methods, such as A-Star (A*), use heuristic functions to systematically search for the shortest path. They guarantee deterministic solutions and are widely applied in structured grid-based settings. However, their scalability remains a challenge, and they often require significant computational resources in high-dimensional or unstructured environments [9], [10]. Hybrid methods combine sampling and optimization to improve path planning performance [11], [12]. For example, hybrid A* integrates heuristic optimization into sampling techniques, enabling it to refine paths with higher precision [13]. This makes it suitable for applications that demand smooth trajectories or strict motion constraints. However, the increased computational complexity reduces its efficiency in high-dimensional or cluttered 3D environments [14]. Reinforcement learning-based methods, on the other hand, rely on extensive training to learn navigation policies. These methods can adapt to dynamic and uncertain environments but require large datasets and high computational power. This dependency limits their scalability and real-time applicability [15], [16].

Neural motion planning methods leverage deep learning to predict feasible paths and adapt to complex environments. Their reliance on pre-trained models and large datasets can restrict their generalization to previously unseen scenarios. Furthermore, their adaptability to changing environmental conditions remains limited, while their high computational demands pose challenges for real-time path planning [17], [18]. In particular, the RRT* algorithm is well known because it can progressively generate paths that increasingly approach the optimal solution, i.e., that path generated by sampling a sufficiently large number of points exhaustively approaches the optimal solution [19]. This feature allows the RRT* algorithm to keep some certain flexibility in different environments and has obvious superiority in high dimensional planning problems [20]. Unfortunately, the RRT* algorithm is limited. For instance, its random sampling strategy should be more uniform. Specifically, when there are many obstacles or narrow passages in an environment, this can slow the initial path generation [21]. In addition, the algorithm lacks its built-in path-smoothing algorithm. However, additional post-processing steps are normally needed to improve the path quality to satisfy practical application needs [22]. Moreover, to update the path tree rapidly, the RRT algorithm requires assistance in dynamic or rapidly changing environments. Second, as a result of its limited adaptability, the overall robustness of the algorithm is affected as well [23]. Two problems are discussed which illustrate the need to work on computational efficiency, path quality, and adaptability to dynamic environments when improving sampling path planning algorithms. On the other hand, only by striking a balance between these aspects, can we try to meet the practical application requirements in the complex obstacle environment [24].

To overcome the limitations of the traditional RRT* algorithm, many researchers have offered different improved algorithms. The improvements presented improve sampling efficiency, generate more optimal paths, and increase obstacle avoidance capability. The initial path generation process is accelerated through the use of a dual-tree expansion method. In low-dimensional spaces, this approach is efficient [25]. Informed -RRT uses a heuristic sampling approach that concentrates on the optimal path, dramatically increasing path optimization and planning efficiencies [26]. The attraction and repulsion models of the artificial potential field are combined in the APF-RRT*. It provides clear directives to the path and makes avoiding obstacles easier. This algorithm does well in environments with fewer obstacles [27]. Yet, existing algorithms are constrained in complex and dynamic settings. When the obstacle is dense, the Bidirectional RRT's dual-tree expansion method dramatically drops off its efficiency [28]; Informed RRT depends heavily on the heuristic function and may easily get trapped in local optima in dynamic environments, which compromises its path planning performance [29]; APF-RRT* degrades rapidly in planning efficiency in areas where the obstacle is dense because the attractive and repulsive forces interfere too much [30].

A widely known method is Biased Random Sampling. It provides improved sampling efficiency and improves the path. Instead, it favours sampling in particular regions, making good use of sampling on certain regions and filtering out random sampling in areas with many obstacles. The heuristic-based biased sampling methods are shown to be effective. In addition, they quickly find high-potential regions in the search space and significantly accelerate the generation of initial paths [31]. However, the method also has limitations in practical applications. For instance, prior knowledge is also needed in defining the biased region. In addition, it is difficult to adjust the sampling strategy in a real-time manner in dynamic environments, which might lead to a lack of global search capability [32]. To overcome these obstacles, researchers have suggested a dynamic sampling strategy. The distribution of the sampling points is adjusted according to real-time environmental changes using these methods, which improves the capability and robustness of the algorithm [33]. While dynamic sampling strategies can improve planning efficiency, their computational cost is still high for high-dimensional spaces. Moreover, the sampling randomness may lead to inconsistent path quality [34].

The artificial Potential Field (APF) model is with simplicity and efficiency widely adopted in path planning algorithms. Obstacles are repulsive sources and the target position is an attractive source. It quickly produces obstacle obstacle-avoiding paths [35]. However, the classic APF model comes with two main limitations. The first problem is that in high obstacle densities, attractive and repulsive forces that interact with each other can lead the path to get stuck in local optima. Further, the parameters of the potential field are prone to the path generation effect, and if the parameters are not properly

tuned it may cause algorithm failure in path planning [36], which calls for research in improvement methods, including the incorporation of a dynamic optimization mechanism into parameters of potential field or the application of a multi-level potential field model to alleviate the failure [37]. While this has reduced these issues somewhat, we have yet to enhance their performance in extremely complex dynamic environments [38].

This manuscript strives to overcome the limitations of existing algorithms and further presents an improved path planning algorithm DPF-Bi-RRT*, which can greatly increase the path generation efficiency and the quality of paths generated, as well as the field adaptability in complex situations. The Biased Random Sampling strategy, in the first place, dynamically updates the sampling point distribution and focuses on exploring high-potential regions [39]. The sample efficiency and the path diversity can be improved through this method [40]. Second, the Dynamic Goal Biasing mechanism updates the sampling probability of the target based on the real-time state of the planning process. In other words, this approach successfully increases the global search ability of the algorithm and further enhances its flexibility in dynamically changing environments [41]. Next, the Improved Artificial Potential Field model contains dual attractive sources and a dual-layer repulsive model. DPF-Bi-RRT* improves efficiency over initial path generation, path quality and adaptability when operating in complex environments, smoothing the path and greatly improving local obstacle avoidance capabilities [42]. The new theoretical foundation and technical support for AAV path planning in complex dynamic environments are provided.

The main contributions of this study can be summarized as follows: Firstly, the Biased Random Sampling strategy greatly improves path generation efficiency, outperforming in densely obstacle environments; Secondly, the Dynamic Goal Biasing strategy enhances the global search capability and adaptability to dynamic environments. This allows the algorithm to achieve high efficiency in solving problems which are real-world complex and dynamically changing. Third, the Improved Artificial Potential Field model is implemented to optimize path smoothness and enhance obstacle avoidance, improving the path planning quality greatly. New theoretical foundations and technical support for autonomous AAV navigation in complex dynamic environments are offered by these innovations.

The structure of This manuscript is as follows: The second part clarifies the definition of the path planning problem and models the related background. The third section is about summing up the research progress of existing path-planning algorithms and analyzing their limitations. The fourth section gives a detailed description of the design and implementation of DPF RRT*. In the fifth section, experiments are done to validate the performance of the algorithm and a comparative analysis with other algorithms is given. The last section of the research work is the concluding section.

II. PROBLEM FORMULATION

In the context of path planning, the entire environment is defined as a configuration space X , where the regions occupied by obstacles are denoted as X_{obs} and the free space is represented as X_{free} such that $X_{free} = X \setminus X_{obs}$. The goal of path planning is to find a continuous path $\sigma: [0, 1] \rightarrow X_{free}$ from a starting point $x_{start} \in X_{free}$ to a target point $x_{goal} \in X_{free}$ ensuring that every point along the path lies within the free space. In other words, $\sigma(t) \in X_{free}, \forall t \in [0, 1]$.

Definition 1 (Feasible Path Planning): This involves finding any feasible path that satisfies collision-free constraints. The path planning problem is defined as (1).

$$\sigma(0) = x_{start}, \sigma(1) = x_{goal}, \sigma(t) \in X_{free}, \forall t \in [0, 1] \quad (1)$$

If such a path exists, the planning is considered successful; otherwise, it fails.

Definition 2 (Optimal Path Planning): That is, finding the optimal path from all feasible paths limiting on minimum cost function, as shown in (2). Limiting on minimum cost function while searching on the goal.

$$\sigma^* = \underset{\sigma \in \Sigma_{feasible}}{\operatorname{argmin}} c(\sigma) \quad (2)$$

The target function is cost function $c(\sigma)$ over paths σ , usually set to some function of path length or travel time. This manuscript adopts two criteria for performance evaluations of path-planning algorithms. In practice, we guarantee that when the number of samples approaches infinity, the algorithm will find a feasible path with probability one, as shown in (3). Assuming the number of samples is infinity, the cost of the path found by the algorithm converges to the globally optimal solution which is given, as shown in (4)

$$\lim_{n \rightarrow \infty} P(\Sigma_{feasible} \neq \emptyset) = 1 \quad (3)$$

$$\lim_{n \rightarrow \infty} c(\sigma^*) = c^* \quad (4)$$

III. RELATED WORK

In this section, we give a detailed introduction to the RRT* algorithm, which serves as the theoretical foundation of the proposed DPF-Bi-RRT* algorithm. Bi-RRT* routines and Artificial potential field coupled Bi-RRT* (APF-Bi-RRT*). The goal of these analyses is to provide a firm theoretical basis for these proposed innovations. This research work compares the strengths and weaknesses of these algorithms, then, based on logical and theoretical analysis, demonstrates the strength and logic behind the innovation proposed in Section IV.

A. RRT*

On the other hand, the asymptotically optimal path planning algorithm RRT* is a classic. The main goal is iterative improvement of path quality by optimizing the structure of the path tree [43], [44]. The outline of the core steps is as follows: Sampling, Expansion, Selection of Parent Node and Path Rewiring. In algorithm 1, we show the main workflow of the algorithm.

Algorithm 1 RRT*

```

Input  $x_{\text{start}}, x_{\text{goal}}, X_{\text{obs}}, \text{Iter}_{\text{max}}$ 
1  $V \leftarrow \{x_{\text{start}}\}, E \leftarrow \emptyset$ 
2 for  $i = 1$  to  $\text{Iter}_{\text{max}}$  do
3    $x_{\text{rand}} \leftarrow \text{Sample}(i)$ 
4    $x_{\text{nearest}} \leftarrow \text{NearestNode}(V, x_{\text{rand}})$ 
5    $x_{\text{new}} \leftarrow \text{Steer}(x_{\text{nearest}}, x_{\text{rand}})$ 
6   if  $\text{NoCollision}(x_{\text{nearest}}, x_{\text{new}}, X_{\text{obs}})$  then
7      $x_{\text{parent}} \leftarrow \text{ChooseParent}(x_{\text{new}}, X_{\text{near}})$ 
8      $V \leftarrow V \cup \{x_{\text{new}}\}, E \leftarrow E \cup \{x_{\text{parent}}, x_{\text{new}}\}$ 
9      $\text{Rewire}(V, E, x_{\text{new}}, X_{\text{near}})$ 
10  end if
11 end for
12 Return  $G = (V, E)$ 

```

The algorithm starts out with the use of the function Sample to produce a random sampling point x_{rand} . Now, the NearestNode function finds the nearest node x_{nearest} from the current path tree to x_{rand} . Then, the Steer function creates a new node x_{new} which is a transition from x_{nearest} to x_{rand} . If the NoCollision function is confident that the path is collision-free then x_{new} is added to the path tree.

RRT* uses the ChooseParent function to minimally increment cost during the path expansion process, selecting the best parent node for x_{new} . The ChooseParent function shown in Algorithm 2, decides based on the candidate parent nodes in the neighborhood computed using their path cost and chooses the best node as the parent of x_{new} . Besides, the overall cost of a path tree is also reduced, and in RRT*, the Rewire function is applied to check whether we can further reduce the cost of the path from other nodes nearby x_{new} . The Rewire function, as shown in Algorithm 3, rewires the connections of neighbouring nodes to optimize the structure of path tree and thus guarantee the asymptotic optimality of the global path.

Algorithm 2 ChooseParent

```

1:  $x_{\text{parent}} \leftarrow x_{\text{nearest}}$ 
2: for each  $x_{\text{near}} \in X_{\text{near}}$  do
3:   if  $\text{NoCollision}(x_{\text{near}}, x_{\text{new}}, X_{\text{obs}})$  then
4:   if  $C(x_{\text{near}}) + \text{Dist}(x_{\text{near}}, x_{\text{new}}) < C(x_{\text{parent}})$  then
5:      $x_{\text{parent}} \leftarrow x_{\text{near}}$ 
6:   end if
7: end if
8: end for
9: Return:  $x_{\text{parent}}$ 

```

While RRT* achieves good path quality optimization, there are several weaknesses to this algorithm as well. For example, uniform sampling frequently leads to a decrease in efficiency near the goal regions, which hinders convergence of the path tree towards regions with high optimal values. Furthermore, the computational complexity of the Rewire function increases as the number of nodes increases, making it difficult to achieve good efficiency of the overall algorithm.

Algorithm 3 Rewire

```

1: Input:  $V, E, x_{\text{new}}, X_{\text{near}}$ 
2: for each  $x_{\text{near}} \in X_{\text{near}}$  do
3:   if  $\text{NoCollision}(x_{\text{near}}, x_{\text{new}}, X_{\text{obs}})$  then
4:     if  $C(x_{\text{new}}) + \text{Dist}(x_{\text{new}}, x_{\text{near}}) < C(x_{\text{near}})$  then
5:        $x_{\text{parent}} \leftarrow x_{\text{new}}$ 
6:        $E \leftarrow (E - (x_{\text{old\_parent}}, x_{\text{near}})) \cup (x_{\text{new}}, x_{\text{near}})$ 
7:     end if
8:   end if
9: end for
10: Return  $G = (V, E)$ 

```

These limitations are analyzed more deeply to provide the critical insights that are proposed in the following work for considering more efficient path-planning algorithms.

The details of implementation and logic are explained in detail below: The set of neighbour nodes within a certain radius of the newly generated node x_{new} is denoted as the parameter X_{near} . Potential connection links are evaluated with this set to minimize path cost. The cumulative path cost from the starting node x_{start} to any node x is denoted by the cost function $C(x)$, which is the key parameter for the quality path judgment. Obvious example of this would be the function $\text{Dist}(x_1, x_2)$ expressing the Euclidean distance between the nodes x_1 and x_2 , so that connecting x_{new} with its neighboring nodes is feasible. Furthermore, x_{parent} has a child equal to x . During the optimization process itself, the connection whose path cost is the minimum is dynamically updated. The reflected parameter E is a set of edges of the path tree, which is updated dynamically during the algorithm, so that the path tree retains the structure of the original tree.

B. BI-RRT*

The RRT* algorithm is extended to Bidirectional RRT*. Simultaneously constructing two trees, one from start and one from the goal improves path plan efficiency [45], [46]. The algorithm expands two trees at separate times and tries to connect them so that we have a complete path in each iteration [47], [48]. The Bidirectional RRT* grows part of its solution to and from the goal, effectively reducing the amount of search time while improving the quality of the planned path. Algorithm 4 describes the core process of the algorithm.

Several key parts are the core steps of the Bi-RRT* algorithm. At first, the start tree T_{start} and the goal tree T_{goal} are initialized such that first tree T_{start} only consists of the start node x_{start} and the second tree T_{goal} only of the goal node x_{goal} . Next, inside each iteration the algorithm dynamically chooses an expansion tree and a connection tree by calling $\text{SelectTrees}(T_{\text{start}}, T_{\text{goal}})$. In other words, the expansion tree creates new nodes and tries to connect to the nodes so created by the connection tree. Random sampling points are used to identify the nearest node $\text{NearestNode}(T_a, x_{\text{rand}})$ and generate a new node x_{new} as $\text{Steer}(x_{\text{nearest}}, x_{\text{rand}})$. Current node is the newly added node, and the generated path is collision-free so it adds the new node to the

Algorithm 4 Bi-RRT*

```

1: Initialize  $T_{\text{start}} \leftarrow \{x_{\text{start}}\}$ ,  $T_{\text{goal}} \leftarrow \{x_{\text{goal}}\}$ 
2: for  $i \leftarrow 1$  to Iter do
3:    $\{T_a, T_b\} \leftarrow \text{SelectTrees}(T_{\text{start}}, T_{\text{goal}})$ 
4:    $x_{\text{rand}} \leftarrow \text{SampleFree}(X_{\text{obs}})$ 
5:    $x_{\text{nearest}} \leftarrow \text{NearestNode}(T_a, x_{\text{rand}})$ 
6:    $x_{\text{new}} \leftarrow \text{Steer}(x_{\text{nearest}}, x_{\text{rand}})$ 
7:   if  $\text{CollisionFree}(x_{\text{nearest}}, x_{\text{new}}, X_{\text{obs}})$  then
8:      $T_a \leftarrow \text{AddNode}(T_a, x_{\text{new}})$ 
9:      $T_a \leftarrow \text{ChooseParent}(T_a, x_{\text{new}})$ 
10:     $T_a \leftarrow \text{Rewire}(T_a, x_{\text{new}})$ 
11:    if  $\text{Connect}(T_a, T_b)$  then
12:      Path  $\leftarrow \text{ExtractPath}(T_{\text{start}}, T_{\text{goal}})$ 
13:      Return Path
14:    end if
15:  end if
16: end for
17: Return "No Path Found"

```

expansion tree, then, it attempts to connect the two trees using $\text{Connect}(T_a, T_b)$.

The results showed that the Bidirectional RRT* algorithm consistently improved the efficiency and quality of the path planning, but it still suffers from limitations in complex environments. For example, its sampling process is often unoptimized for near-obstacle paths, resulting in suboptimal local paths [49], [50]. Moreover, the computational complexity increases due to the dynamic bidirectional expansion of the trees.

T_{start} and T_{goal} are start tree and goal tree in the Bi RRT algorithm. At first, these trees have only one start point in x_{start} and one goal point in x_{goal} . To optimize search efficiency, the algorithm chooses, at each stage, the expansion tree T_a and the connection tree T_b dynamically. The function SelectTrees is then used to implement this selection strategy. The expansion tree T_a gets presupposed to expand and the function AddNode adds a new node x_{new} to T_a in such a way that there is connectivity between them. The function Connect is called whenever T_a and T_b try to connect and helps the two create connections between the endpoint nodes of its trees. In case the connection succeeds, the algorithm uses ExtractPath to get the full path and return as the final planning result [51], [52], [53].

C. APF-Bi-RRT*

APF-Bi-RRT* is a new path-planning method that combines APF principles into Bi-RRT* with a bidirectional extension mechanism [54], [55]. Using the strengths of APF, this algorithm attempts to keep the advantages of Bi-RRT* to enhance guidance and efficiency of expansion while preserving the capabilities of Bi-RRT* for exploring complex environments. The incorporation of attractive and repulsive forces makes APF-Bi-RRT* adaptively change the expansion directions of the two trees [56], leading to more

efficient obstacle avoidance and smoother path generation. Algorithm 5 shown below is a detailed description of the algorithm process.

Algorithm 5 APF-Bi-RRT*

```

1:  $T_{\text{start}} \leftarrow \{x_{\text{start}}\}$ ,  $T_{\text{goal}} \leftarrow \{x_{\text{goal}}\}$ ,  $E_{\text{start}}, E_{\text{goal}} \leftarrow \emptyset$ 
2: for  $i \leftarrow 1$  to Iter do
3:    $T_a, T_b \leftarrow \text{SelectTrees}(T_{\text{start}}, T_{\text{goal}})$ 
4:    $x_{\text{rand}} \leftarrow \text{Sample}(X_{\text{free}})$ 
5:    $x_{\text{nearest}} \leftarrow \text{NearestNode}(T_a, x_{\text{rand}})$ 
6:    $x_{\text{new}} \leftarrow \text{Steer}(x_{\text{nearest}}, x_{\text{rand}})$ 
7:    $F_{\text{total}} \leftarrow F_{\text{att}}(x_{\text{new}}) + F_{\text{rep}}(x_{\text{new}})$ 
8:   Adjust  $x_{\text{new}}$  based on  $F_{\text{total}}$ 
9:   if  $\text{NoCollision}(x_{\text{nearest}}, x_{\text{new}}, X_{\text{obs}})$  then
10:    Add  $x_{\text{new}}$  to  $T_a$  and  $E_a$ 
11:    if  $\text{Connect}(T_a, T_b)$  then
12:      Extract the final path and exit
13:    end if
14:     $x_{\text{parent}} \leftarrow \text{ChooseParent}(x_{\text{new}}, X_{\text{near}})$ 
15:    Rewire  $(T_a, x_{\text{new}}, X_{\text{near}})$ 
16:  end if
17: end for
18: Output: Path or Failure

```

The algorithm calculates the total artificial potential force F_{total} at x_{new} , which is a combination of attractive and repulsive forces. These forces are dynamically adjusted based on the proximity to obstacles and the goal. Using F_{total} , the position of x_{new} is refined to improve obstacle avoidance. If the adjusted x_{new} does not result in collision (verified by the NoCollision function), it is added to T_a . The algorithm then attempts to connect T_a and T_b through the connect function. Finally, the algorithm updates the tree structure using the ChooseParent and Rewire functions to optimize the path cost and ensure global efficiency.

The attractive force (F_{att}) pulls the path toward the target, while the repulsive force (F_{rep}) helps to avoid obstacles. These two forces combine dynamically into a total force (F_{total}) that guides the expansion direction. The repulsive force acts within a specific range, defined by d_{rep} , which ensures effective obstacle avoidance. The coefficients k_{att} and k_{rep} adjust the relative strength of attraction and repulsion. E_{start} and E_{goal} maintain the connectivity of the start and goal trees. During the bidirectional search, T_a and T_b represent the active and passive trees.

D. LIMITATIONS AND MOTIVATION

Existing path planning algorithms are still much constrained in terms of efficiency to generate a path and in quality of the generated path and in the adaptation to complex environments [57], [58]. Because the RRT* algorithm offers asymptotic optimality, its uniform random sampling tends to degrade local search efficiency in such complex scenarios. In particular, this issue gets worse around the target regions, as it takes too long to converge to better paths around the

targets [59], [60]. A dual tree extension strategy is used in bidirectional RRT* to improve path generation speed. While it does not include an effective guidance mechanism for tree selection and node connection, poor global search efficiency and redundant paths are frequently produced [61]. At the same time, APF-Bi-RRT* introduces artificial potential fields to improve path quality, while its potential field model is sensitive to local optima in dynamic and densely obstructive environments. In fact, this sensitivity may lead to oscillation problems in the vicinity of obstacles.

To mitigate the limitations of existing algorithms, DPF-Bi-RRT* is designed with key innovations: region biased random sampling, which dynamically biases the goal, and an enhanced artificial potential field model. The improvements aim to fix all three deficiencies of current path generation efficiency, path optimization quality, and adaptability to complex environments.

IV. DPF-BI-RRT*

A. DYNAMIC GOAL BIASING STRATEGY

Dynamic Goal Biasing Strategy addresses the limitations of fixed goal sampling probability in traditional RRT* algorithms [62]. The approach dynamically adjusts the goal bias probability P_{bias} based on the obstacle density near the sampling region. Figure 1 illustrates the algorithm's principle, showing how the goal bias probability adapts to dense and sparse obstacle distributions to optimize path planning.

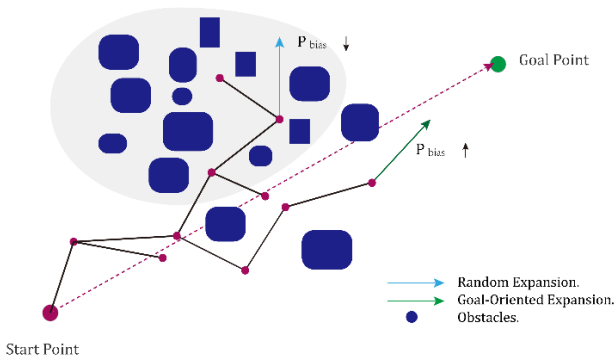


FIGURE 1. Path expansion behavior under dynamic goal biasing strategy. P_{bias} , ↓: Reduce goal sampling probability; P_{bias} , ↑: Increase goal sampling probability.

The obstacle density ρ is a key parameter influencing the dynamic adjustment of P_{bias} . It is calculated by using (5).

$$\rho = \frac{V_{\text{obs}}}{\frac{4}{3}\pi d_{\text{rep}}^3} \quad (5)$$

where V_{obs} represents the volume of obstacles within the detection range d_{rep} . A higher obstacle density indicates a more complex region. Using ρ , P_{bias} is dynamically adjusted as shown in (6).

$$P_{\text{bias}} = \frac{\exp(-\beta_1 \rho)}{\beta_2} \quad (6)$$

Here, β_1 controls the influence of ρ on P_{bias} and β_2 normalizes the probability. This adjustment enables the algorithm to focus on goal-oriented expansion in sparse regions while promoting random exploration in dense regions.

The direction of path expansion depends on the comparison between $\text{rand}()$ and P_{bias} . If $\text{rand}() < P_{\text{bias}}$, the path extends towards the goal, as calculated by (7). Otherwise, it expands towards a random point, as described in (8).

$$\mathbf{P}_{\text{new}} = \mathbf{P} + \Delta s \cdot \frac{\mathbf{g} - \mathbf{P}}{\|\mathbf{g} - \mathbf{P}\|}, \quad \text{if } \text{rand}() < P_{\text{bias}} \quad (7)$$

$$\mathbf{P}_{\text{new}} = \mathbf{P} + \Delta s \cdot \frac{\mathbf{r}_{\text{rand}} - \mathbf{P}}{\|\mathbf{r}_{\text{rand}} - \mathbf{P}\|}, \quad \text{otherwise} \quad (8)$$

where $\mathbf{P}_{\text{new}}$ is the new expansion point, $\mathbf{P}$ is the current point, $\mathbf{g}$ is the goal, Δs is the step size, and $\mathbf{r}_{\text{rand}}$ is a random sample point. Using (7) and (8), the algorithm achieves a balance between local optimization and global exploration by dynamically adjusting the path expansion direction. Algorithm 6 outlines the detailed steps of the Dynamic Goal Biasing Strategy. It integrates the above principles to enhance adaptability and efficiency in complex environments. The pseudocode is presented in a tabular format for clarity.

Algorithm 6 Dynamic Goal Biasing Strategy

```

1: Input:  $p, g, r_{\text{rand}}$ .
2: Output: New position  $p_{\text{new}}$ .
3: if  $\text{rand}() < P_{\text{bias}}$  then
4:    $p_{\text{new}} \leftarrow p + \Delta s \cdot \frac{g - p}{\|g - p\|}$ 
5: else
6:    $p_{\text{new}} \leftarrow p + \Delta s \cdot \frac{r_{\text{rand}} - p}{\|r_{\text{rand}} - p\|}$ 
7: end if
8: Return:  $p_{\text{new}}$ .
```

B. BIASED RANDOM SAMPLING STRATEGY

To overcome the limitations of traditional RRT* algorithms in complex environments, such as low global search efficiency and limited path quality, a Biased Random Sampling strategy is proposed [63]. This strategy dynamically divides the search space and adjusts sampling probabilities to concentrate sampling points in high-efficiency regions of the target path [64]. By integrating global path information and local environmental complexity, it prioritizes exploration of potentially efficient path regions. This balances search efficiency and global optimization capability. The conceptual framework of this strategy is illustrated in Figure 2, while the detailed procedure is outlined in Algorithm 7.

The search space is first divided into uniform cubic grids based on dimensions $x_{\text{max}}, y_{\text{max}}, z_{\text{max}}$, and the grid size l . The total number of grids, N_{total} , is calculated using (9), where $x_{\text{bins}} = \lceil \frac{x_{\text{max}}}{l} \rceil$, $y_{\text{bins}} = \lceil \frac{y_{\text{max}}}{l} \rceil$, and $z_{\text{bins}} = \lceil \frac{z_{\text{max}}}{l} \rceil$. Each grid center c_i is evaluated based on its proximity to the target path, calculated as shown in (10).

$$N_{\text{total}} = x_{\text{bins}} \cdot y_{\text{bins}} \cdot z_{\text{bins}} \quad (9)$$

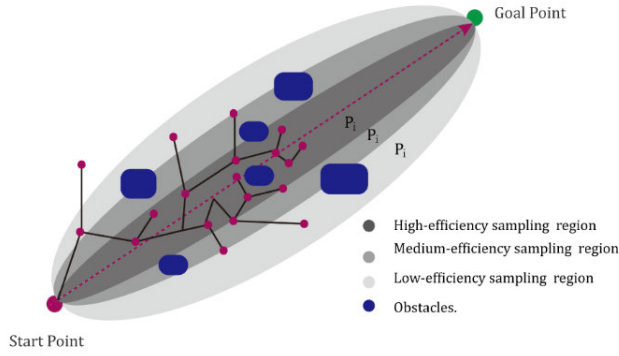


FIGURE 2. Illustration of biased random sampling strategy with dynamic grid partitioning. P_i : Sampling probability, P_i decreases as the color gradient becomes lighter.

Algorithm 7 Biased Random Sampling Strategy

```

1: Input:  $x_{\max}, y_{\max}, z_{\max}, l, \sigma$ .
2:  $x_{\text{bins}} \leftarrow \lceil x_{\max}/l \rceil, y_{\text{bins}} \leftarrow \lceil y_{\max}/l \rceil, z_{\text{bins}} \leftarrow \lceil z_{\max}/l \rceil$ .
3:  $N_{\text{total}} \leftarrow x_{\text{bins}} \cdot y_{\text{bins}} \cdot z_{\text{bins}}$ .
4: For  $c_i \in \text{Grid Centers}$ :
5:    $d_{\text{line}} = \frac{|g-s| - |c_i-s|}{|g-s|}$ .
6:    $S_i \leftarrow \exp\left(-\frac{d_{\text{line}}^2}{2\sigma^2}\right)$ .
7:  $P_i \leftarrow S_i / \sum_{j=1}^{N_{\text{total}}} S_j$ .
8:  $c_{\text{rand}} \leftarrow \text{RandomGrid}(P_i)$ .
9:  $r_{\text{rand}} \leftarrow c_{\text{rand}} + \delta \cdot l, \delta \sim U(-0.5, 0.5)$ .
10: end for
11: Output:  $r_{\text{rand}}$ .

```

$$d_{\text{line}} = \frac{|g-s| - |c_i-s|}{|g-s|} \quad (10)$$

where s and g denote the start and goal points, respectively. d_{line} measures the alignment of c_i with the target path.

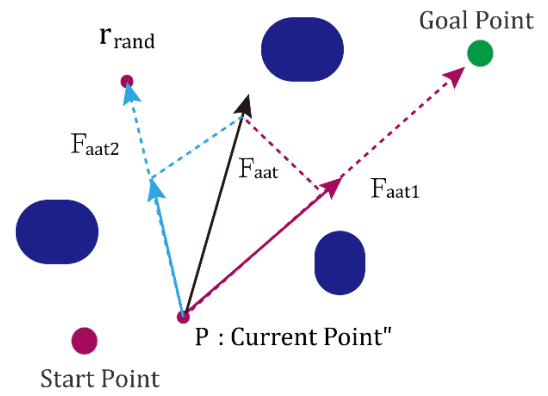
A distribution weight S_i is assigned to each grid center using a Gaussian function in (11), where σ determines the spread of the distribution. The parameter σ is defined based on the Euclidean distance between the start and goal points, as shown in (12). where g and s represent the goal and start points, respectively. The spread of the Gaussian function is proportional to the problem's scale, ensuring adaptive distribution across the search space. After normalization, the sampling probability for each grid is computed using (13). A random grid center c_{rand} is selected according to P_i , and within this grid, the final random sampling point is determined using (14).

$$S_i = \exp\left(-\frac{d_{\text{line}}^2}{2\sigma^2}\right) \quad (11)$$

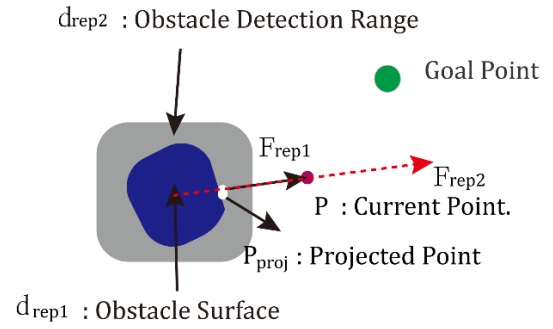
$$\sigma = 0.5 \times \|g-s\| \quad (12)$$

$$P_i = \frac{S_i}{\sum_{j=1}^{N_{\text{total}}} S_j} \quad (13)$$

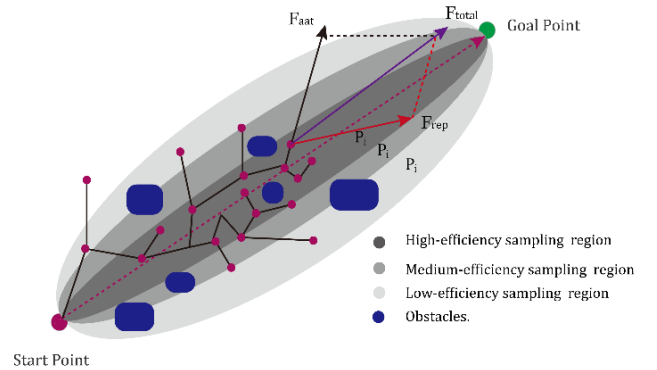
$$r_{\text{rand}} = c_{\text{rand}} + \delta \cdot l \quad (14)$$



(a) Dual Attraction Sources



(b) Dual Repulsion Sources



(c) Resultant Force and Path Expansion

FIGURE 3. Dual attraction and dual repulsion models in the improved artificial potential field.

where $\delta = [\delta_x, \delta_y, \delta_z] \sim U(-0.5, 0.5)$ ensures the randomness of the sample location within the grid.

C. IMPROVED ARTIFICIAL POTENTIAL FIELD

The APF method is a common approach in local path planning, modeling the goal point as an attractive source and obstacles as repulsive sources [65]. To address the limitations of traditional APF in dense obstacle environments, this study proposes an improved method incorporating dual-attraction and dual-repulsion models. Figure 3 illustrates the core principle, and Algorithm 8 presents the implementation steps.

Algorithm 8 Improved Artificial Potential Field

```

1:   Input:  $p, g, r_{\text{rand}}, d_{\text{rep}}, k_{\text{att}}, k_{\text{rand}}, k_{\text{rep}}, n$ .
2:   Output:  $d_{\text{apf}}$ .
3:    $F_{\text{att}} \leftarrow -k_{\text{att}}(p - g) - k_{\text{rand}}(p - r_{\text{rand}})$ .
4:   if  $d \leq d_{\text{rep}}$  then
5:      $F_{\text{rep1}} \leftarrow k_{\text{rep}} \left( \frac{1}{d} - \frac{1}{d_{\text{rep}}} \right) \frac{1}{d^2} (p - p_{\text{proj}})$ .
6:   else
7:      $F_{\text{rep1}} \leftarrow 0$ .
8:   end if
9:    $F_{\text{rep2}} \leftarrow \frac{n}{2} k_{\text{rep}} \left( \frac{1}{d} - \frac{1}{d_{\text{rep}}} \right)^2 (p - g)$ .
10:   $F_{\text{rep}} \leftarrow F_{\text{rep1}} + F_{\text{rep2}}$ .
11:   $F_{\text{total}} \leftarrow F_{\text{att}} + F_{\text{rep}}$ .
12:   $d_{\text{apf}} \leftarrow \frac{F_{\text{total}}}{\|F_{\text{total}}\|}$ .
13:  Return:  $d_{\text{apf}}$ .

```

The attractive force F_{att} combines the influences of the goal point g and a random target point r_{rand} . As shown in (15), where k_{att} and k_{rand} are coefficients that control the attraction strengths of the goal point and the random target point, respectively. The dual-repulsion model introduces two layers of repulsive forces. The first layer, F_{rep1} , focuses on the distance between the current node p and the obstacle's surface p_{proj} . The second layer, F_{rep2} , introduces a gradient-based repulsive force, are shown in (16) and (17). The total repulsive force F_{rep} combines these two layers, as shown in (18).

$$F_{\text{att}} = -k_{\text{att}}(p - g) - k_{\text{rand}}(p - r_{\text{rand}}) \quad (15)$$

$$F_{\text{rep1}} = \begin{cases} k_{\text{rep}} \left(\frac{1}{d} - \frac{1}{d_{\text{rep}}} \right) \frac{1}{d^2} (p - p_{\text{proj}}), & d \leq d_{\text{rep}} \\ 0, & d > d_{\text{rep}} \end{cases} \quad (16)$$

$$F_{\text{rep2}} = \begin{cases} \frac{n}{2} k_{\text{rep}} \left(\frac{1}{d} - \frac{1}{d_{\text{rep}}} \right)^2 (p - g), & d \leq d_{\text{rep}} \\ 0, & d > d_{\text{rep}} \end{cases} \quad (17)$$

$$F_{\text{rep}} = F_{\text{rep1}} + F_{\text{rep2}} \quad (18)$$

Finally, the total force is calculated by combining the attractive and repulsive forces, as shown in (19). To determine the direction for path extension, the total force is normalized as (20). Equation (15)-(20) describe the computation of the attractive, repulsive, and total forces, as well as the normalized path extension direction. The dual-attraction model enhances global guidance, while the dual-repulsion model ensures precise obstacle avoidance. Together, they improve the adaptability and robustness of the path planning algorithm in complex environments.

$$F_{\text{total}} = F_{\text{att}} + F_{\text{rep}} \quad (19)$$

$$d_{\text{apf}} = \frac{F_{\text{total}}}{\|F_{\text{total}}\|} \quad (20)$$

V. SIMULATION RESULTS

The simulation environment is set in a three-dimensional space with dimensions $200 \times 200 \times 100$ units. The start



FIGURE 4. BIM model of the commercial and residential project in Fuquan, Keqiao, Shaoxing, Zhejiang Province, China.

and goal points are fixed at $[190, 190, 50]$ and $[20, 20, 20]$. Obstacles are modeled as high-rise buildings with varying heights to emulate urban conditions [66]. Three environments with increasing complexity are designed: Environment A has no obstacles, Environment B contains eight moderately dense obstacles, and Environment C includes nineteen obstacles arranged to replicate a dense urban layout. Figure 4 shows the BIM-based model of a commercial and residential project in Fuquan, Keqiao, Shaoxing, Zhejiang Province, China, which serves as a reference for Environment C. Each simulation is run 100 times to ensure reliable results. Four algorithms are tested: RRT*, Bi-RRT*, APF-Bi-RRT*, and DPF-Bi-RRT*. Key parameters include a step size of 7 units, a goal tolerance of 5 units, and a rewire radius of 10 units. The evaluation metrics are runtime, number of nodes, and path length. All simulations are performed in MATLAB.

A. ENVIRONMENT A

In Environment A, the simulation space is set without any obstacles, representing an ideal scenario for AAV trajectory planning. The algorithms RRT*, Bi-RRT*, APF-Bi-RRT*, and DPF-Bi-RRT* are applied to the same start and goal points in the three-dimensional space. Figure 5 illustrates the simulation results for the four algorithms, showing the generated paths in the absence of obstacles. Figure 5(a) displays the trajectory planned by RRT*, characterized by dense node sampling and an overall longer path. Figure 5(b) represents Bi-RRT*, which demonstrates improved efficiency due to its bidirectional nature but still generates a longer path compared to the hybrid methods. Figure 5(c) shows APF-Bi-RRT*, where the path is smoother and shorter, benefiting from the artificial potential field's guidance. Figure 5(d) presents DPF-Bi-RRT*, which achieves the shortest and most optimized trajectory with significantly fewer nodes.

The quantitative results of the algorithms in Environment A are summarized in table 1. RRT* required the highest average number of nodes (339.04) and achieved a success rate of 98%. Bi-RRT* and APF-Bi-RRT* both reached a 100%

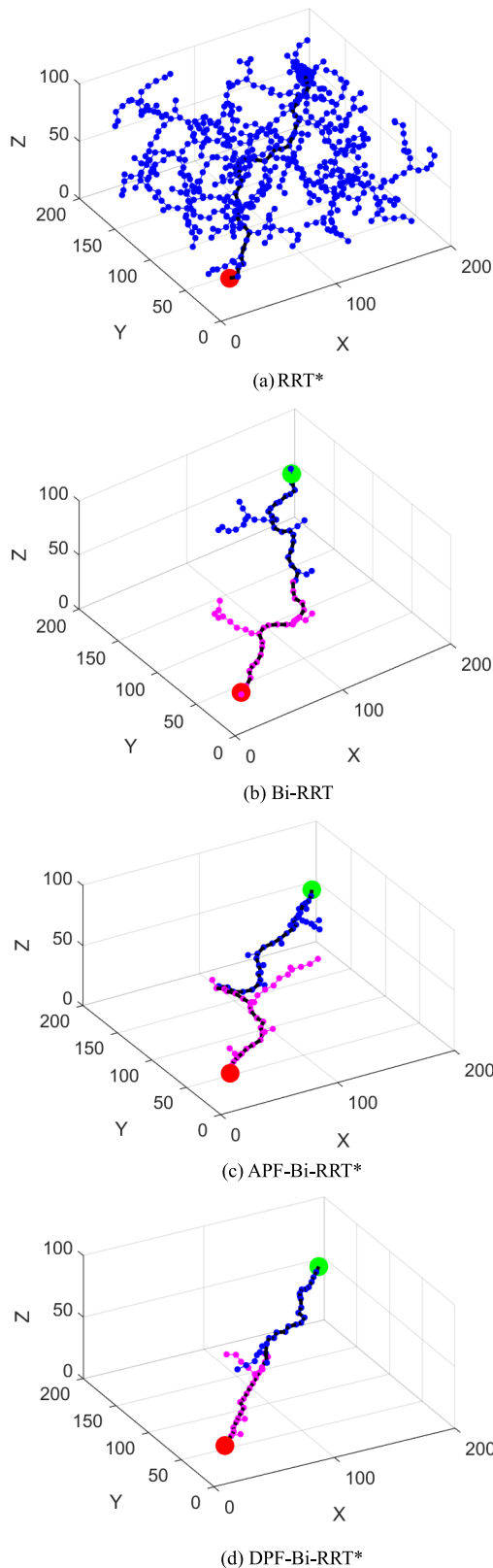


FIGURE 5. Simulation in Environment A.

success rate, with average node counts of 116.51 and 107.76, respectively. However, the most efficient algorithm was

TABLE 1. Quantitative results comparison of 4 algorithms in environment with no obstacles.

Algorithm	Average Nodes	Average Processing Time (second)	Average Length (meter)	Success Rate
RRT*	339.0408	0.7399	310.5716	98%
Bi-RRT*	116.5100	0.8193	326.5500	100%
APF-Bi-RRT*	107.7600	0.7609	295.2488	100%
DPF-Bi-RRT*	56.1500	0.1592	264.7186	100%

TABLE 2. Quantitative results comparison of 4 algorithms in environment with 8 obstacles.

Algorithm	Average Nodes	Average Processing Time (second)	Average Length (meter)	Success Rate
RRT*	327.5474	0.8722	327.0484	95%
Bi-RRT*	125.3200	1.5859	331.3091	100%
APF-Bi-RRT*	119.9000	0.9815	304.9463	100%
DPF-Bi-RRT*	74.5100	0.3864	281.3572	100%

DPF-Bi-RRT*, which reduced the average node count to 56.15 and achieved the shortest path length (264.72) with an average runtime of only 0.1592 seconds. These results confirm that DPF-Bi-RRT* significantly outperforms other algorithms in terms of efficiency and trajectory optimization in a simple, obstacle-free environment.

B. ENVIRONMENT B

In Environment B, the simulation includes eight obstacles, represented as high-rise buildings distributed across the three-dimensional space. The algorithms RRT*, Bi-RRT*, APF-Bi-RRT*, and DPF-Bi-RRT* are tested to assess their performance in navigating around these obstacles. Figure. 6(a) shows the RRT* algorithm generating a trajectory with a dense distribution of nodes around obstacles, resulting in longer paths. Figure. 6(b) illustrates Bi-RRT*, which uses bidirectional searching to improve efficiency but still struggles with obstacle avoidance. Figure. 6(c) demonstrates APF-Bi-RRT*, where the artificial potential field component helps guide the trajectory away from obstacles, resulting in a smoother path. Figure. 6(d) highlights DPF-Bi-RRT*, which achieves the shortest and most efficient trajectory, minimizing node density even in the presence of obstacles.

The simulation results for Environment B are presented in table 2. RRT* shows the highest average node count (327.55) and the longest path length (327.05), with a success rate of 95%. Bi-RRT* and APF-Bi-RRT* both achieved 100% success rates, with node counts of 125.32 and 119.90, respectively, and reduced path lengths. DPF-Bi-RRT* significantly outperformed all other algorithms, achieving the shortest path length (281.36), the lowest node count (74.51), and the

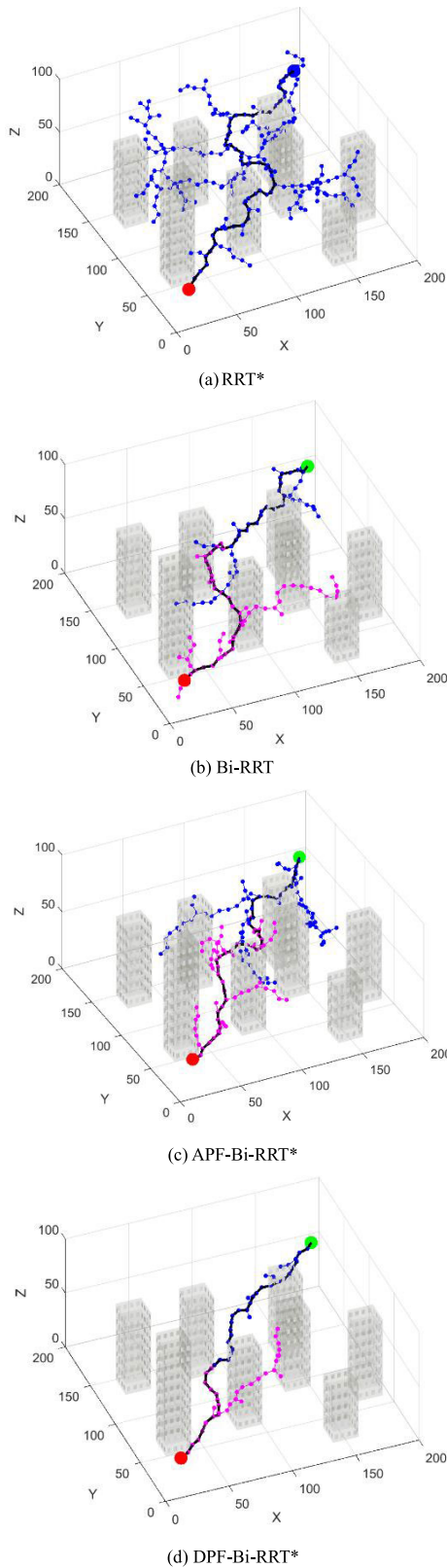


FIGURE 6. Simulation in Environment B.

fastest average runtime (0.3864 seconds). These results confirm that DPF-Bi-RRT* demonstrates superior performance in environments with moderate obstacle density.

TABLE 3. Quantitative results comparison of 4 algorithms in environment with 19 obstacles.

Algorithm	Average Nodes	Average Processing Time (second)	Average Length (meter)	Success Rate
RRT*	292.9011	0.9667	340.6435	91%
Bi-RRT*	134.4200	2.1928	357.5546	100%
APF-Bi-RRT*	128.5300	1.3963	325.0339	100%
DPF-Bi-RRT*	103.6000	0.9778	308.1447	100%

C. ENVIRONMENT C

Environment C contains 19 high-rise buildings with varying heights, designed to match the building density of a commercial and residential project in Fuquan, Keqiao, Shaoxing, Zhejiang Province, China. The environment represents the structure of a modern city, providing a realistic scenario for AAV trajectory planning. Figure 7(a) illustrates the RRT* algorithm, which produces a trajectory with dense node sampling and frequent detours, resulting in suboptimal paths. Figure 7(b) demonstrates Bi-RRT*, where bidirectional search improves efficiency, but the algorithm struggles to maintain path optimization in such a dense obstacle configuration. Figure 7(c) presents APF-Bi-RRT*, where the artificial potential field component helps navigate the AAV around obstacles, generating smoother trajectories. Figure 7(d) highlights DPF-Bi-RRT*, which achieves an efficient and optimized path with minimal node density, even in a highly cluttered environment.

The performance metrics for Environment C are summarized in table 3. RRT* achieves the lowest success rate (91%) and generates the highest average path length (340.64) and node count (292.90). Bi-RRT* and APF-Bi-RRT* achieve 100% success rates, with average node counts of 134.42 and 128.53, respectively. However, the DPF-Bi-RRT* algorithm outperforms all others in this challenging scenario, achieving the shortest path length (308.14), the lowest node count (103.60), and maintaining an average runtime of 0.9778 seconds. These results highlight the robustness of DPF-Bi-RRT* in handling complex environments with dense obstacle configurations.

D. PATH OPTIMIZATION

The path optimization process uses cubic spline interpolation to smooth the initially computed paths in the three-dimensional space. The input consists of path nodes $P = [p_1, p_2, \dots, p_n]$, where each node $p_i = (x_i, y_i, z_i)$ represents a discrete point along the trajectory. For each dimension x , y , and z , a cubic spline interpolation is performed to create continuous and smooth trajectories. The interpolation generates denser nodes $P_{opt} = [q_1, q_2, \dots, q_m]$, where $m \gg n$, providing a higher resolution path.

The optimized path is checked for collisions with obstacles. Each segment is analyzed, and if any segment collides,

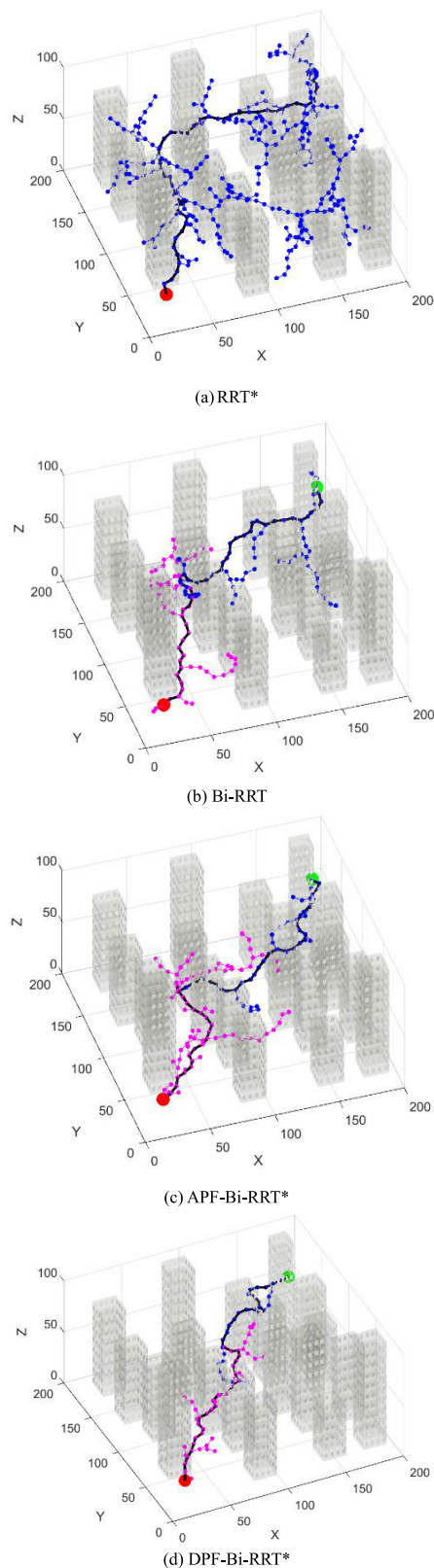


FIGURE 7. Simulation in Environment C.

adjustments are made or parts of the path are reverted to avoid obstacles. This ensures that the smoothed path

TABLE 4. Average nodes in three environments.

Average Nodes	Environment A	Environment B	Environment C
RRT*	339	328	293
Bi-RRT*	117	125	134
APF-Bi-RRT*	108	120	129
DPF-Bi-RRT*	56	75	104

TABLE 5. Average processing time (second) in three environments.

Average Time	Environment A	Environment B	Environment C
RRT*	0.7	0.9	1.0
Bi-RRT*	0.8	1.6	2.2
APF-Bi-RRT*	0.8	1.0	1.4
DPF-Bi-RRT*	0.2	0.4	1.0

TABLE 6. Average path length in three environments.

Average Length	Environment A	Environment B	Environment C
RRT*	311	327	341
Bi-RRT*	327	331	358
APF-Bi-RRT*	295	305	325
DPF-Bi-RRT*	265	281	308

remains feasible in the environment. This optimization process improves path smoothness and reduces abrupt directional changes. It enhances path quality while ensuring safety in three-dimensional environments with dense obstacles. The resulting optimized paths are better suited for real-time navigation.

E. PARAMETER SENSITIVITY ANALYSIS AND DISCUSSION

The data in tables 4, 5, and 6 provide a detailed comparison of the performance of the four algorithms across three distinct environments. DPF-Bi-RRT* demonstrates consistent efficiency in all metrics compared to RRT*, Bi-RRT*, and APF-Bi-RRT*.

Table 4 shows DPF-Bi-RRT* uses significantly fewer nodes than other algorithms, reflecting superior sampling efficiency. For example, in Environment A, it expands only 56 nodes compared to RRT*'s 339. This efficiency is maintained in more complex scenarios, with minimal increases in node usage across all environments. APF-Bi-RRT* also shows reduced node counts but does not match the consistent reductions achieved by DPF-Bi-RRT*. Bi-RRT* falls in between, with moderate node reduction. In Table 5, DPF-Bi-RRT* achieves the shortest runtime in all three environments, especially in Environment A, where its runtime is only

TABLE 7. Pearson's correlation table.

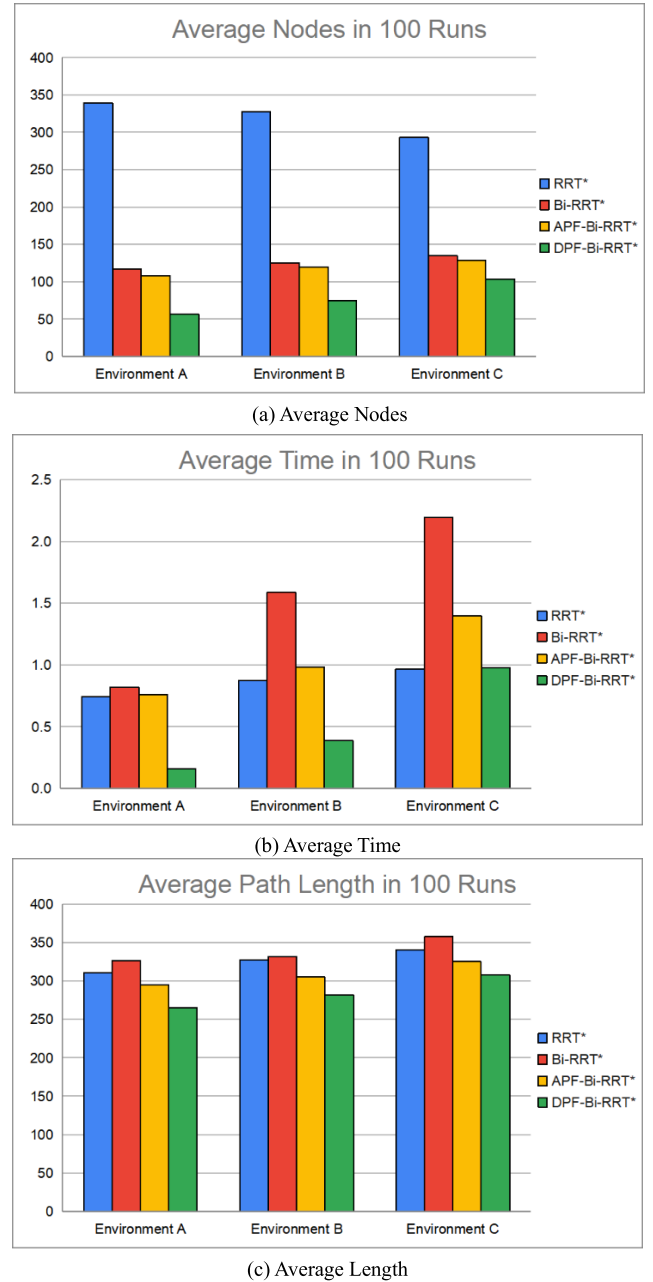
Parameter	Average Nodes	Average Length	Average Time
β_1	0.8737	0.9012	0.6372
β_2	0.8060	0.7111	0.6282
K_{att}	0.5303	-0.9466	0.5489
K_{rep}	0.1317	0.9030	0.1202
K_{rand}	-0.2580	0.9557	-0.5025
d_{rep}	-0.9386	-0.9541	-0.8796

0.2 seconds. Its performance remains competitive even in Environment C, matching RRT*'s runtime while significantly outperforming Bi-RRT* and APF-Bi-RRT*. Bi-RRT* exhibits increasing runtimes as environment complexity grows, highlighting limitations in managing computational efficiency. Table 6 shows that DPF-Bi-RRT* consistently produces the shortest paths, indicating enhanced optimization capabilities. Its paths are markedly shorter than those of RRT*, with differences of up to 46 units in Environment A. APF-Bi-RRT* generates shorter paths than Bi-RRT* and RRT*, but its improvement is less consistent compared to DPF-Bi-RRT*. Bi-RRT* often produces longer paths, particularly in complex environments, suggesting limited adaptability in path refinement.

Figure 8 illustrates these findings visually, highlighting the advantages of DPF-Bi-RRT* across all metrics. Its ability to achieve efficient node usage, low computation times, and short paths demonstrates robustness and adaptability to varying environmental complexities. These features make it a more suitable choice for efficient and reliable path planning in 3D environments.

To investigate the impact of different parameters on algorithm performance, Table 7 further examines the impact of key parameters on the algorithm's performance. Among them, d_{rep} exhibits the strongest correlation with all three evaluation metrics: node count (-0.9386), path length (-0.9541), and computation time (-0.8796). However, since d_{rep} corresponds to the sensor's detection range, it is inherently limited by hardware capabilities. Adjusting this parameter in simulations may not generalize well to real-world applications, as its value is constrained by the AAV's onboard sensor specifications. Therefore, we exclude d_{rep} from further sensitivity analysis and focus on the influence of β_1 , β_2 , K_{att} , K_{rep} , and K_{rand} .

The tested parameter values are as follows: β_1 [0.25, 0.4, 0.5, 0.6, 0.75, 1], β_2 [1, 1.1, 1.25, 1.5, 2, 3], K_{att} [1, 2, 2.25, 2.5, 2.75, 3, 4], K_{rep} [0.3, 0.4, 0.5, 0.75, 1, 1.25], and K_{rand} [3, 3.5, 4, 4.5, 5, 6]. Each parameter was tested across 50 independent simulation runs, totaling 1800 simulations. Results show that β_1 and β_2 exhibit strong positive correlations with both average node count (0.8737, 0.8060) and path length (0.9012, 0.7111), indicating that higher

**FIGURE 8.** Algorithm performance in 100 runs.

values lead to increased sampling density and longer paths. K_{att} strongly influences path length (-0.9466), suggesting that higher attraction forces shorten paths but slightly increase node count (0.5303). The effects of K_{rep} and K_{rand} on path optimization differ: K_{rep} has a positive correlation with path length (0.9030), indicating that stronger repulsion forces lead to longer paths, while K_{rand} shows a strong positive correlation with path length (0.9557) but a weak negative correlation with node count (-0.2580), meaning higher randomness enhances exploration. These findings highlight the importance of tuning parameters to balance path quality and computational efficiency.

VI. CONCLUSION

We propose the DPF-Bi-RRT* algorithm, a novel three dimensional path planning algorithm based on combining a dual potential field mechanism with a biased random sampling strategy. It solves challenges created by cluttered and dynamic environments by integrating to help both global path optimization and local obstacle avoidance. They balance global guidance with precise local obstacle avoidance dual according to processional dual-attraction and dual-repulsion mechanisms thus adaptability in complex scenarios is guaranteed. An efficient sampling distribution is yielded by the Biased Random Sampling Strategy by concentrating computational effort on high potential path regions thereby reducing overhead significantly without compromising solution quality. The algorithm is compared to RRT*, Bi-RRT*, and APF-Bi-RRT* with comparative simulations across different environments showing the effectiveness of the method.

The DPF-Bi-RRT* algorithm proves to improve significantly in computationally efficiency, node reduction, and path length, and is shown to find collision free, smooth, and globally optimal paths. Taking into account its reduced computational cost and improved adaptability, the algorithm is especially suitable for AAV navigation under real time conditions when rapid decisions and high accuracy are needed. The proposed algorithm presents a practical answer for real world applications including urban AAV delivery, disaster response, and autonomous exploration through its ability to deal with varying environmental complexities and to generate optimal paths. It is more efficient so quicker routes can be generated for emergencies where rapid deployment is necessary. Additionally, the approach facilitates new advancements, such as real time sensing of the environment, handling of dynamical obstacles, and adaptation to behavior in uncertain terrains. The DPF-Bi-RRT* introduces an efficient, robust, and scalable methodology to the evolving problem of AAV path planning, allowing promising applications in both research and reality.

REFERENCES

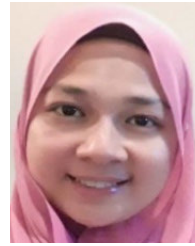
- [1] G. Hu, M. Cheng, E. H. Houssein, and H. Jia, "CMPSO: A novel co-evolutionary multigroup particle swarm optimization for multi-mission UAVs path planning," *Adv. Eng. Informat.*, vol. 63, Jan. 2025, Art. no. 102923.
- [2] S. K. Phang, T. H. A. Chiang, A. Happonen, and M. M. L. Chang, "From satellite to UAV-based remote sensing: A review on precision agriculture," *IEEE Access*, vol. 11, pp. 127057–127076, 2023.
- [3] A. Mancini, E. Frontoni, and P. Zingaretti, "Satellite and UAV data for precision agriculture applications," in *Proc. Int. Conf. Unmanned Aircr. Syst. (ICUAS)*, Jun. 2019, pp. 491–497.
- [4] L. Ge, S. K. Phang, and J. Che, "Intelligent monitoring with unmanned aerial vehicles for advanced concrete replacement studies," *J. Eng. Sci. Technol.*, vol. 7, pp. 153–166, Jan. 2024.
- [5] S. Boubaker and L. Kolsi, "Improved Dijkstra algorithm for mobile robot path planning and obstacle avoidance," *Comput., Mater. Continua*, vol. 72, pp. 5939–5954, Jan. 2022.
- [6] A. Orthey, C. Chamzas, and L. E. Kavraki, "Sampling-based motion planning: A comparative review," *Annu. Rev. Control, Robot., Auto. Syst.*, vol. 7, no. 1, pp. 285–310, Jul. 2024.
- [7] J. D. Gammell and M. P. Strub, "Asymptotically optimal sampling-based motion planning methods," *Annu. Rev. Control, Robot., Auto. Syst.*, vol. 4, no. 1, pp. 295–318, May 2021.
- [8] G. Chen, N. Luo, D. Liu, Z. Zhao, and C. Liang, "Path planning for manipulators based on an improved probabilistic roadmap method," *Robot. Comput.-Integr. Manuf.*, vol. 72, Dec. 2021, Art. no. 102196.
- [9] K. Jang, J. Baek, S. Park, and J. Park, "Motion planning for closed-chain constraints based on probabilistic roadmap with improved connectivity," *IEEE/ASME Trans. Mechatronics*, vol. 27, no. 4, pp. 2035–2043, Aug. 2022.
- [10] M. N. Ab Wahab, A. Nazir, A. Khalil, W. J. Ho, M. F. Akbar, M. H. M. Noor, and A. S. A. Mohamed, "Improved genetic algorithm for mobile robot path planning in static environments," *Expert Syst. Appl.*, vol. 249, Sep. 2024, Art. no. 123762.
- [11] X. Yu, N. Jiang, X. Wang, and M. Li, "A hybrid algorithm based on grey wolf optimizer and differential evolution for UAV path planning," *Expert Syst. Appl.*, vol. 215, Apr. 2023, Art. no. 119327.
- [12] J. Li, Y. Huang, J. Wu, X. Wang, and Z. Ning, "Energy efficiency maximization for STAR-RIS and UAV-assisted IUA: A multi-agent DRL approach," *IEEE Internet Things J.*, early access, Dec. 16, 2024, doi: 10.1109/JIOT.2024.3518081.
- [13] Z. Zhang, J. Jiang, J. Wu, and X. Zhu, "Efficient and optimal penetration path planning for stealth unmanned aerial vehicle using minimal radar cross-section tactics and modified A-star algorithm," *ISA Trans.*, vol. 134, pp. 42–57, Mar. 2023.
- [14] P. Cao, L. Lei, S. Cai, G. Shen, X. Liu, X. Wang, L. Zhang, L. Zhou, and M. Guizani, "Computational intelligence algorithms for UAV swarm networking and collaboration: A comprehensive survey and future directions," *IEEE Commun. Surveys Tuts.*, vol. 26, no. 4, pp. 2684–2728, 4th Quart., 2024.
- [15] W. D. Lukito, W. Xiang, P. Lai, P. Cheng, C. Liu, and K. Yu, "Integrated STAR-RIS and UAV for satellite IoT communications: An energy-efficient approach," *IEEE Internet Things J.*, early access, Oct. 1, 2024, doi: 10.1109/JIOT.2024.3472019.
- [16] Z. Yu, Z. Si, X. Li, D. Wang, and H. Song, "A novel hybrid particle swarm optimization algorithm for path planning of UAVs," *IEEE Internet Things J.*, vol. 9, no. 22, pp. 22547–22558, Nov. 2022.
- [17] H. Shiri, J. Park, and M. Bennis, "Remote UAV online path planning via neural network-based opportunistic control," *IEEE Wireless Commun. Lett.*, vol. 9, no. 6, pp. 861–865, Jun. 2020.
- [18] H. U. Zijian, G. A. O. Xiaoguang, W. A. N. Kaifang, Z. H. A. I. Yiwei, and W. A. N. G. Qianglong, "Relevant experience learning: A deep reinforcement learning method for UAV autonomous motion planning in complex unknown environments," *Chin. J. Aeronaut.*, vol. 34, no. 12, pp. 187–204, 2021.
- [19] I.-B. Jeong, S.-J. Lee, and J.-H. Kim, "Quick-RRT: Triangular inequality-based implementation of RRT with improved initial solution and convergence rate," *Expert Syst. Appl.*, vol. 123, pp. 82–90, Jun. 2019.
- [20] H. Huang, Y. Shang, X. Liu, X. Liu, and P. Qi, "An improved Bi-RRT-based path planning algorithm with adaptive search strategy assignment mechanism for ultra-low-altitude penetration of fixed-wing aircraft," *Aerosp. Sci. Technol.*, vol. 152, Sep. 2024, Art. no. 109363.
- [21] I. Noreen, A. Khan, and Z. Habib, "Optimal path planning using RRT based approaches: A survey and future directions," *Int. J. Adv. Comput. Sci. Appl.*, vol. 7, no. 11, pp. 97–104, 2016.
- [22] L. Jiang, S. Liu, Y. Cui, and H. Jiang, "Path planning for robotic manipulator in complex multi-obstacle environment based on improved RRT," *IEEE/ASME Trans. Mechatronics*, vol. 27, no. 6, pp. 4774–4785, Dec. 2022.
- [23] C. Jiang, Z. Hu, Z. P. Mourelatos, D. Gorsich, P. Jayakumar, Y. Fu, and M. Majcher, "R2-RRT: Reliability-based robust mission planning of off-road autonomous ground vehicle under uncertain terrain environment," *IEEE Trans. Autom. Sci. Eng.*, vol. 19, no. 2, pp. 1030–1046, Apr. 2022.
- [24] Y. Wang, X. Li, X. Zhuang, F. Li, and Y. Liang, "A sampling-based distributed exploration method for UAV cluster in unknown environments," *Drones*, vol. 7, no. 4, p. 246, Apr. 2023.
- [25] J. Guo, W. Xia, X. Hu, and H. Ma, "Feedback RRT algorithm for UAV path planning in a hostile environment," *Comput. Ind. Eng.*, vol. 174, Dec. 2022, Art. no. 108771.

- [26] S. K. Phang, S. Lai, F. Wang, M. Lan, and B. M. Chen, "UAV calligraphy," in *Proc. 11th IEEE Int. Conf. Control Autom. (ICCA)*, Jun. 2014, pp. 422–428.
- [27] J. Fan, X. Chen, and X. Liang, "UAV trajectory planning based on bi-directional APF-RRT algorithm with goal-biased," *Expert Syst. Appl.*, vol. 213, Mar. 2023, Art. no. 119137.
- [28] H. Fan, J. Huang, X. Huang, H. Zhu, and H. Su, "BI-RRT: An improved path planning algorithm for secure and trustworthy mobile robots systems," *Heliyon*, vol. 10, no. 5, Mar. 2024, Art. no. e26403.
- [29] S. K. Phang, S. Lai, F. Wang, M. Lan, and B. M. Chen, "Systems design and implementation with jerk-optimized trajectory generation for UAV calligraphy," *Mechatronics*, vol. 30, pp. 65–75, Sep. 2015.
- [30] D. Wu, L. Wei, G. Wang, L. Tian, and G. Dai, "APF-IRRT: An improved informed rapidly-exploring random trees-star algorithm by introducing artificial potential field method for mobile robot path planning," *Appl. Sci.*, vol. 12, no. 21, p. 10905, Oct. 2022.
- [31] T. Kattenborn, J. Lopatin, M. Förster, A. C. Braun, and F. E. Fassnacht, "UAV data as alternative to field sampling to map woody invasive species based on combined Sentinel-1 and Sentinel-2 data," *Remote Sens. Environ.*, vol. 227, pp. 61–73, Jun. 2019.
- [32] S. Puliti, L. T. Ene, T. Gobakken, and E. Næsset, "Use of partial-coverage UAV data in sampling for large scale forest inventories," *Remote Sens. Environ.*, vol. 194, pp. 115–126, Jun. 2017.
- [33] Z. Xu, D. Deng, and K. Shimada, "Autonomous UAV exploration of dynamic environments via incremental sampling and probabilistic roadmap," *IEEE Robot. Autom. Lett.*, vol. 6, no. 2, pp. 2729–2736, Apr. 2021.
- [34] J. Qi, H. Yang, and H. Sun, "MOD-RRT: A sampling-based algorithm for robot path planning in dynamic environment," *IEEE Trans. Ind. Electron.*, vol. 68, no. 8, pp. 7244–7251, Aug. 2021.
- [35] Y. Zhao, K. Liu, G. Lu, Y. Hu, and S. Yuan, "Path planning of UAV delivery based on improved APF-RRT algorithm," in *Proc. J. Phys., Conf.*, Oct. 2020, vol. 1624, no. 4, Art. no. 042004.
- [36] L. Zhang, N. An, and Z. Ma, "Research of hybrid path planning with improved A and TEB in static and dynamic environments," *J. Supercomput.*, vol. 180, pp. 18009–18047, May 2024.
- [37] X. Meng and X. Fang, "A UGV path planning algorithm based on improved A with improved artificial potential field," *Electronics*, vol. 13, no. 5, p. 972, Mar. 2024.
- [38] L. Zhao, Y. Liu, and S. Chen, "High-dimensional path optimization for UAVs in dynamic environments," *J. Control Theory Appl.*, vol. 29, no. 1, pp. 120–132, Jan. 2023.
- [39] H. Gao, X. Hou, J. Xu, and B. Guan, "Quad-rotor unmanned aerial vehicle path planning based on the target bias extension and dynamic step size RRT algorithm," *World Electr. Vehicle J.*, vol. 15, no. 1, p. 29, 2024.
- [40] H. Qu, C. Zheng, H. Ji, K. Barai, and Y.-J. Zhang, "A fast and efficient approach to estimate wild blueberry yield using machine learning with drone photography: Flight altitude, sampling method and model effects," *Comput. Electron. Agricult.*, vol. 216, Jan. 2024, Art. no. 108543.
- [41] A. M. Smith, B. T. Johnson, and C. A. Brown, "Dynamic environment path planning for UAVs using artificial potential fields," *Int. J. Robot. Res.*, vol. 39, no. 12, pp. 1456–1474, Dec. 2020.
- [42] W. Zhao, A. Tan, and C. Ren, "An innovative path planning algorithm for complex obstacle environments with adaptive obstacle density adjustment: AODA-PF-RRT," *Electronics*, vol. 13, no. 20, p. 4047, Oct. 2024.
- [43] S. M. LaValle and J. J. Kuffner, "Rapidly-exploring random trees: Progress and prospects: Steven M. Lavalle, Iowa state University, a James j. Kuffner, jr., University of Tokyo, Tokyo, Japan," in *Algorithmic and Computational Robotics*. New York, NY, USA: CRC Press, 2001, pp. 303–307, doi: 10.1201/9781439864135.
- [44] J. Wang, W. Chi, C. Li, C. Wang, and M. Q.-H. Meng, "Neural RRT: Learning-based optimal path planning," *IEEE Trans. Autom. Sci. Eng.*, vol. 17, no. 4, pp. 1748–1758, Oct. 2020.
- [45] Z. Tahir, A. H. Qureshi, Y. Ayaz, and R. Nawaz, "Potentially guided bidirectionalized RRT for fast optimal path planning in cluttered environments," *Robot. Auto. Syst.*, vol. 108, pp. 13–27, Oct. 2018.
- [46] L. Wang, Z. Wang, Z. Ying, X. Bai, and N. Xu, "A path planning framework based on an improved weighted heuristic RRT and optimization strategy," *IEEE Trans. Intell. Vehicles*, vol. 9, no. 1, pp. 1941–1952, Jan. 2024.
- [47] L. Zhang, H. Yin, A. Li, L. Hu, and L. Duo, "RSA-RRT: A path planning algorithm based on restricted sampling area," *J. King Saud Univ.-Comput. Inf. Sci.*, vol. 36, no. 8, Oct. 2024, Art. no. 102152.
- [48] Y. Liu, C. Li, H. Yu, and C. Song, "NT-ARS-RRT: A novel non-threshold adaptive region sampling RRT algorithm for path planning," *J. King Saud Univ.-Comput. Inf. Sci.*, vol. 35, no. 9, Oct. 2023, Art. no. 101753.
- [49] C. Yuan, G. Liu, W. Zhang, and X. Pan, "An efficient RRT cache method in dynamic environments for path planning," *Robot. Auto. Syst.*, vol. 131, Sep. 2020, Art. no. 103595.
- [50] J. J. Tai, S. K. Phang, and F. Y. M. Wong, "COAA—An optimized obstacle avoidance and navigational algorithm for UAVs operating in partially observable 2D environments," *Unmanned Syst.*, vol. 10, no. 2, pp. 159–174, 2022.
- [51] H. Tu, Y. Deng, Q. Li, M. Song, and X. Zheng, "Improved RRT global path planning algorithm based on bridge test," *Robot. Auto. Syst.*, vol. 171, Jan. 2024, Art. no. 104570.
- [52] C. Zhang, L. Zhang, L. Zhu, T. Zhang, Z. Xiao, and X.-G. Xia, "3D deployment of multiple UAV-mounted base stations for UAV communications," *IEEE Trans. Commun.*, vol. 69, no. 4, pp. 2473–2488, Apr. 2021.
- [53] Z. Kaleem, W. Khalid, A. Muqabel, A. A. Nasir, C. Yuen, and G. K. Karagiannidis, "Learning-aided UAV 3D placement and power allocation for sum-capacity enhancement under varying altitudes," *IEEE Commun. Lett.*, vol. 26, no. 7, pp. 1633–1637, Jul. 2022.
- [54] Y. Lin and L. Zhang, "An improved quick informed-RRT algorithm based on hybrid bidirectional search and adaptive adjustment strategies," *Intell. Service Robot.*, vol. 17, no. 4, pp. 847–870, May 2024.
- [55] Y. Guo, X. Liu, X. Liu, Y. Yang, and W. Zhang, "FC-RRT: An improved path planning algorithm for UAV in 3D complex environment," *ISPRS Int. J. Geo-Inf.*, vol. 11, no. 2, p. 112, Feb. 2022.
- [56] J. Wang, Z. Zhao, J. Qu, and X. Chen, "APPA-3D: An autonomous 3D path planning algorithm for UAVs in unknown complex environments," *Sci. Rep.*, vol. 14, no. 1, Jan. 2024, Art. no. 1231.
- [57] J. S. Pan, J. X. Lv, L. J. Yan, S. W. Weng, S. C. Chu, and J. K. Xue, "Golden eagle optimizer with double learning strategies for 3D path planning of UAV in power inspection," *Math. Comput. Simul.*, vol. 193, pp. 509–532, Mar. 2022.
- [58] A. Ait Saadi, A. Soukane, Y. Meraihi, A. Benmessaoud Gabis, S. Mirjalili, and A. Ramdane-Cherif, "UAV path planning using optimization approaches: A survey," *Arch. Comput. Methods Eng.*, vol. 29, no. 6, pp. 4233–4284, Oct. 2022.
- [59] J. Muñoz, F. López, F. Quevedo, C. A. Monje, S. Garrido, and L. E. Moreno, "Multi UAV coverage path planning in urban environments," *Sensors*, vol. 21, no. 21, p. 7365, Nov. 2021.
- [60] L. Ge, S. King Phang, and J. Che, "Unmanned aerial vehicle-assisted assessment of cable-stayed bridge slant cables and piers: Methods and applications," in *Proc. Int. Conf. Green Energy, Comput. Sustain. Technol. (GECOST)*, Jan. 2024, pp. 141–145.
- [61] S. Jiang, W. Jiang, and L. Wang, "Unmanned aerial vehicle-based photogrammetric 3D mapping: A survey of techniques, applications, and challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 2, pp. 135–171, Jun. 2022.
- [62] B. Li, Q. Li, Y. Zeng, Y. Rong, and R. Zhang, "3D trajectory optimization for energy-efficient UAV communication: A control design perspective," *IEEE Trans. Wireless Commun.*, vol. 21, no. 6, pp. 4579–4593, Jun. 2022.
- [63] J. J. Tai, S. K. Phang, and Y. M. F. Wong, "Optimized autonomous UAV design with obstacle avoidance capability," in *Proc. AIP Conf.*, 2020, vol. 2233, no. 1, Art. no. 020026.
- [64] M. Maboudi, M. Homaei, S. Song, S. Malihi, M. Saadatseresht, and M. Gerke, "A review on viewpoints and path planning for UAV-based 3-D reconstruction," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 5026–5048, 2023.
- [65] S. Zhang and R. Zhang, "Radio map-based 3D path planning for cellular-connected UAV," *IEEE Trans. Wireless Commun.*, vol. 20, no. 3, pp. 1975–1989, Mar. 2021.
- [66] C. Jianfeng, S. King Phang, and L. Ge, "Optimizing UAV pathways for enhanced 3D architectural reconstruction," in *Proc. Int. Conf. Green Energy, Comput. Sustain. Technol. (GECOST)*, Jan. 2024, pp. 169–173.



LIN GE received the master's degree from the City University of Hong Kong, in 2016. She is currently pursuing the Ph.D. degree with Taylor's University.

She is a Lecturer with the School of Urban Construction, Hangzhou Polytechnic, Zhejiang, China. Since 2017, she has been teaching as a Lecturer. Her research explores AAV swarm path planning and its applications in complex urban environments. Since assuming her academic role, she has actively contributed to teaching and research, authoring and co-authoring several publications in her field. She has presented her work at international conferences and received multiple accolades for her contributions. Her research interests include autonomous systems, urban informatics, and the integration of AAV technology in urban planning. Notably, she was recognized as the Best Presenter at the EURECA International Engineering Conference, in both 2023 and 2024. Additionally, she was awarded the Silver Award for her presentation at the Hyper Interdisciplinary Conference, in 2024.



NOHAIDDA SARIFF received the M.Sc. degree in electrical engineering from Universiti Teknologi MARA (UiTM), Malaysia, in 2011, and the Ph.D. degree in control robotics from Universiti Teknologi Malaysia (UTM), in 2021. She has more than ten years of experience as an Academician. She is currently a Senior Lecturer with the School of Engineering, Faculty of Innovation and Technology, Taylor's University, Malaysia. She has published in several peer-reviewed indexed

journals, conference proceedings, and book chapters. Her current research interests include automation and control robotics, mobile robot navigation, and path-planning algorithms. She is a Registered Professional Engineer with the Board of Engineers Malaysia and a Registered Professional Technologist with the Malaysian Board of Technologists. She has actively participated as a reviewer for both local and international journals.

...



SWEE KING PHANG (Senior Member, IEEE) received the B.Eng. (Hons.) and Ph.D. degrees in electrical engineering from the National University of Singapore (NUS), in 2010 and 2014, respectively. He is currently the Program Director of mechatronics engineering with Taylor's University, Subang Jaya, Malaysia. Prior to joining academia, he was a Research Fellow with NUS and a Research Scientist with Temasek Laboratories, focusing on unmanned systems and control

technologies. He is the Deputy Director of the Digital Innovation and Smart Society Impact Laboratory, Taylor's University, he is actively involved in impactful education initiatives such as the integration of the Impact Laboratory into the university's curriculum. Under his leadership, student teams from his research laboratory have achieved remarkable success, winning competitions such as the Champion title at the AIRBUS Innovation Fun Day in Kuala Lumpur, Malaysia, in 2019, and the CDIO Drone Academy Challenge in Kanazawa, Japan, in 2018. He is a registered Professional Engineer (P.Eng.). He has authored more than 100 publications, including book chapters, peer-reviewed journal articles, and international conference proceedings. His research interests include system design of unmanned aerial systems (UAS), modeling and flight controller design, indoor navigation, and trajectory optimization. He has received several accolades, including the Guan Zhao-Zhi Best Paper Award at the 33rd Chinese Control Conference, in 2013, the Finalist for Best Paper at the 14th IEEE International Conference on Control and Automation, in 2018, and the IET Commendable Paper Award at the 13th International Engineering Research Conference, in 2019.